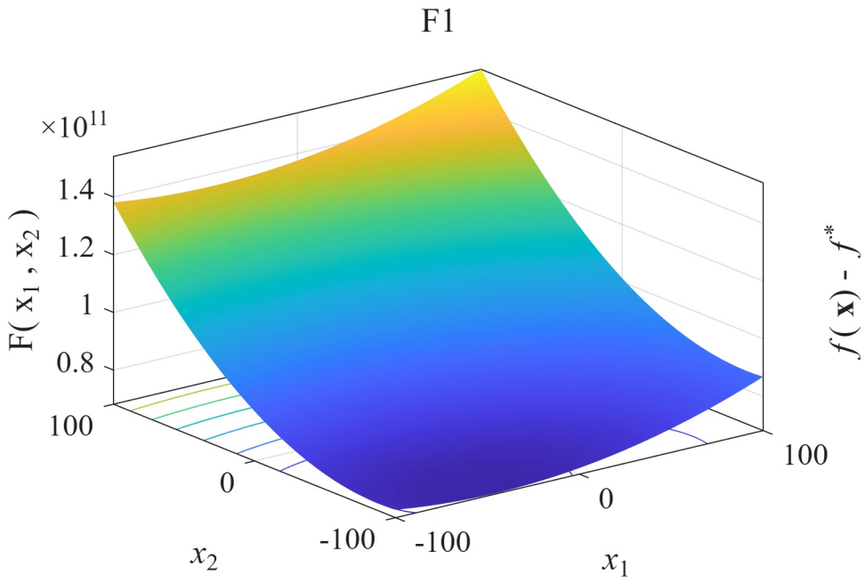
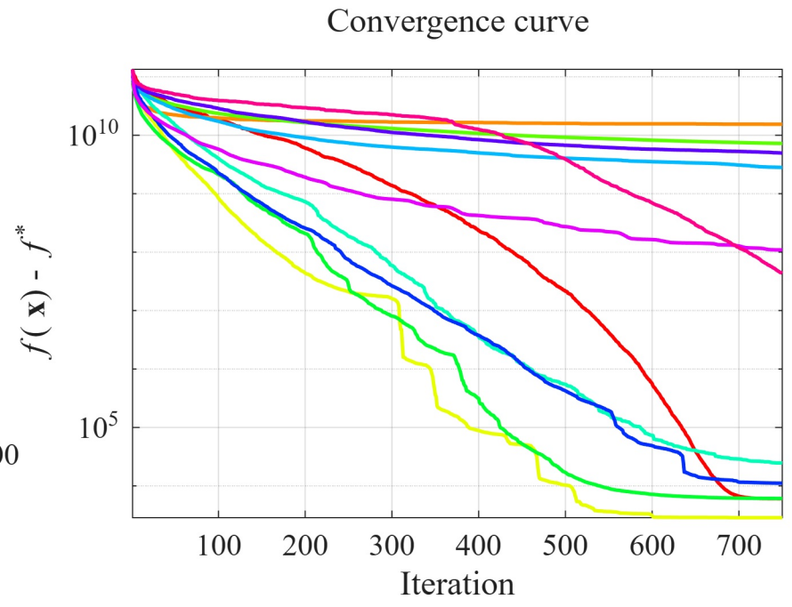
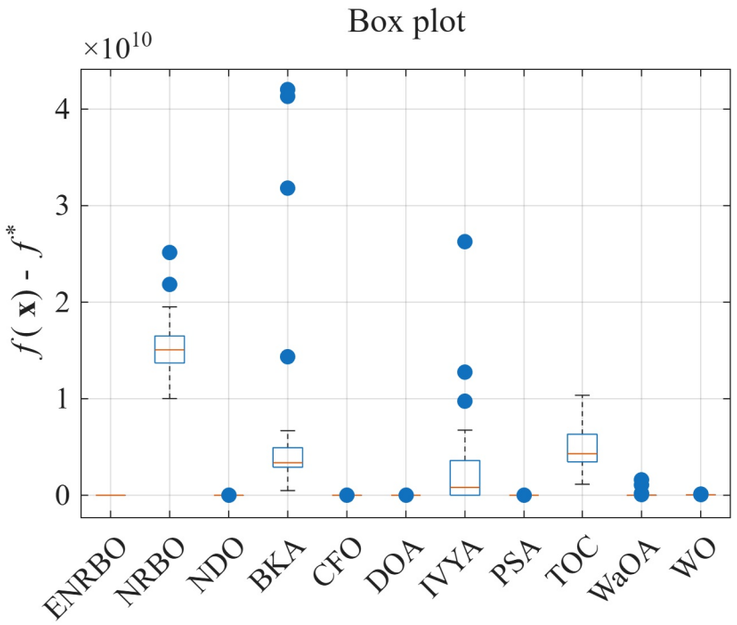

F1
×10^11
1.4
1.2
1
0.8
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1


Convergence curve
10^10
10^5
f(x) - f*
100
200
300
400
500
600
700
Iteration


Box plot
×10^10
4
3
2
1
0
f(x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO

ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO